An Introduction to Pseudopowers

Lecture 3: The Fixed-Point Argument

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Some Easy Results

The Smallest Pseudopower

The following statements are equivalent:

1. $\mathrm{pp}(\lambda) = \lambda^+$.
2. Whenever $\langle \lambda_\alpha : \alpha < \mathrm{cf}(\lambda) \rangle$ is and increasing sequence of regular cardinals cofinal in λ,

$$\mathrm{tcf}\left(\prod_{\alpha<\mathrm{cf}(\lambda)} \lambda_\alpha / J^{bd}\right) = \lambda^+.$$

In other words, every increasing sequence of regular cardinals cofinal in λ carries a scale.

Countably Many Earlier Generators

Suppose $A \subseteq \lambda \cap \mathrm{Reg}$ has order type ω and supremum λ, and

$$\lambda < \kappa \in \mathsf{pcf}(A).$$

If $\mathsf{pcf}(A) \cap (\lambda, \kappa)$ is countable, then some infinite $B \subseteq A$ satisfies

$$\mathsf{tcf}\left(\prod B/\mathrm{Fin}\right) = \kappa.$$

A Small Gap

Suppose $\kappa < \lambda^{+\omega_1}$ is representable at λ of countable cofinality. Then κ is representable at λ modulo the finite ideal.

Below $\lambda^{+\mathfrak{p}}$

The same conclusion holds if κ is representable at λ and $\kappa < \lambda^{+\mathfrak{p}}$.

Larger Supports

Recall $\mathrm{pp}_\theta(\lambda)$ allows supports of size at most θ, with ideals extending the bounded ideal.

If $\mathrm{pp}(\lambda) < \lambda^{+\omega_1}$, then

$$\mathrm{pp}_\theta(\lambda) = \mathrm{pp}(\lambda) \qquad (\aleph_0 \leq \theta < \lambda).$$

In fact, for every unbounded $A \subseteq \lambda \cap \mathrm{Reg}$ with $|A| < \lambda$, some $\sigma < \lambda$ satisfies

$$\max \mathrm{pcf}(A \setminus \sigma) \leq \mathrm{pp}(\lambda).$$

The Same Diagonal Argument

Suppose for every $\sigma < \lambda$ we have

$$\max \mathrm{pcf}(A \setminus \sigma) > \mathrm{pp}(\lambda) \qquad (\sigma < \lambda).$$

Then $\mathrm{pcf}(A) \cap (\lambda, \mathrm{pp}(\lambda)]$ is countable. Diagonalization gives $C \subseteq A$ of order type ω with

$$C \setminus \sigma \notin J_{\leq \mathrm{pp}(\lambda)}[A]$$

for every $\sigma < \lambda$.

An ultrafilter on C avoiding this ideal and all bounded sets therefore gives a representation at λ greater than $\mathrm{pp}(\lambda)$. Contradiction.

Frechet Property

Suppose

$$\mathrm{tcf}\left(\prod B_\kappa/\mathrm{Fin}\right) = \kappa,$$
$$\mathrm{tcf}\left(\prod B_\tau/\mathrm{Fin}\right) = \tau.$$

Then

$$\kappa \neq \tau \quad \Longrightarrow \quad |B_\kappa \cap B_\tau| < \aleph_0.$$

An infinite intersection would inherit both scales.

If $\mathrm{pp}(\lambda) < \lambda^{+\mathfrak{p}}$, we can therefore represent all the representable cardinals at λ modulo finite on pairwise almost disjoint supports.

The (sketchy) main course

Below the Next Fixed Point

Assume λ is strong limit and $\mathsf{cf}(\lambda) = \aleph_0$. Let

$$\chi = \mathsf{min}\{\mu > \lambda : \mu = \aleph_\mu\}.$$

If $\mathsf{pp}(\lambda) < \chi$, then

$$\mathsf{pp}(\lambda) = \mathsf{cov}(\lambda, \lambda, \aleph_1, 2) = \lambda^{\aleph_0}.$$

Or equivalently

If λ is singular strong limit of countable cofinality and $\mathrm{pp}(\lambda) < \mathrm{cov}(\lambda, \lambda, \aleph_1, 2)$ then there is a fixed point $\chi = \aleph_\chi$ such that $\lambda < \chi \leq \mathrm{pp}(\lambda)$.

Our first task is to show that $\kappa_1 < \kappa_3$ implies $\kappa_1 < \kappa_2$. We need to recall something from lecture 2 that I warned you we would need later:

$$\mathsf{cov}(\mathsf{pp}(\lambda), \lambda, \aleph_2, \aleph_1) = \mathsf{pp}(\lambda)$$

So it suffices to prove

$$\kappa_4 \leq \mathsf{cov}(\kappa_2, \lambda, \aleph_2, \aleph_1).$$

Bounding κ_4

We show

$$\kappa_4 \leq Q := \mathsf{cov}(\kappa_2, \lambda, \aleph_2, \aleph_1).$$

Choose a witnessing family

$$\mathcal{C} \subseteq [\kappa_2]^{<\lambda}, \qquad |\mathcal{C}| = Q.$$

Let $\mathcal{A} \subseteq [\kappa_2]^{\aleph_1}$ be strongly almost disjoint:

$$A \neq B \in \mathcal{A} \quad \Longrightarrow \quad |A \cap B| < \aleph_0.$$

Choosing Countable Subsets

Each $A \in \mathcal{A}$ is covered by countably many members of $\mathcal{C}$.

Since A is uncountable, choose

$$C_A \in \mathcal{C}, \qquad b_A \in [A \cap C_A]^{\aleph_0}.$$

Strong almost disjointness makes

$$A \longmapsto b_A$$

injective: an infinite set cannot lie in two distinct members of $\mathcal{A}$.

Counting the Subsets

Since λ is strong limit,

$$\left|[C]^{\aleph_0}\right| < \lambda \qquad (C \in \mathcal{C}).$$

Recall $Q \geq \kappa_2 \geq \mathsf{pp}(\lambda) > \lambda$. Thus

$$|\mathcal{A}| \leq \left|\bigcup_{C \in \mathcal{C}} [C]^{\aleph_0}\right| \leq Q \cdot \lambda = Q.$$

As $\mathcal{A}$ was arbitrary,

$$\kappa_4 \leq \mathsf{cov}(\kappa_2, \lambda, \aleph_2, \aleph_1).$$

First Conclusion

Our assumptions imply $\kappa_1 < \kappa_2$.

We now move on to identifying the fixed point.

The Least Cutoff

Put $\mathfrak{c} = 2^{\aleph_0}$ and choose the least cardinal $\chi \in [\lambda, \mathrm{pp}(\lambda)]$ such that

$$\mathrm{cov}(\mathrm{pp}(\lambda), \chi^+, \mathfrak{c}^+, 2) \leq \mathrm{pp}(\lambda).$$

Such a χ exists as $\chi = \mathrm{pp}(\lambda)$ works for trivial reasons. We will show that χ must be an $\aleph$-fixed point satisfying $\lambda < \chi \leq \mathrm{pp}(\lambda)$.

A promise

We will show $\chi > \lambda$ but postpone the proof for now. Note this a promise to show

$$\mathsf{pp}(\lambda) < \mathsf{cov}(\mathsf{pp}(\lambda), \lambda^+, \mathfrak{c}^+, 2)$$

The Cutoff Is a Fixed Point

We can rule out $\chi < \aleph_\chi$ using the equivalence of cov and pp at non-fixed points:

For some cardinal $\nu \in [\lambda, \chi)$,

$$\mathrm{cov}(\chi, \nu^+, \mathfrak{c}^+, 2) = \chi + \sup_{\substack{\nu < \rho \leq \chi \\ \mathrm{cf}(\rho) \leq \mathfrak{c}}} \mathrm{pp}_{\mathfrak{c}}(\rho).$$

Inverse Monotonicity then bounds every pseudopower on the right by $\mathrm{pp}(\lambda)$, so $\mathrm{cov}(\chi, \nu^+, \mathfrak{c}^+, 2) \leq \mathrm{pp}(\lambda)$.

But...

But we already know $\mathsf{cov}(\mathsf{pp}(\lambda), \chi^+, \mathfrak{c}^+, 2) \leq \mathsf{pp}(\lambda)$. Using $\mathsf{cov}(\chi, \nu^+, \mathfrak{c}^+, 2) \leq \mathsf{pp}(\lambda)$, we can build a family that witnesses

$$\mathsf{cov}(\mathsf{pp}(\lambda), \nu^+, \mathfrak{c}^+, 2) \leq \mathsf{pp}(\lambda).$$

Thus if $\chi > \lambda$ then χ is a fixed point.

(If you work harder, you can show χ must have cofinality least $\aleph_1$. Even more work can improve this to cofinality at least λ^+.)

But why is $\chi > \lambda$?

An Ideal of Sets of Regular Cardinals

Assume by way of contradiction that $\chi = \lambda$ so $\mathsf{cov}(\mathsf{pp}(\lambda), \lambda^+, \mathfrak{c}^+, 2) = \mathsf{pp}(\lambda)$

Define

$$\mathcal{I} = \left\{ A \subseteq \lambda \cap \mathrm{Reg} : \mathsf{cf}_{\mathfrak{c}} \left(\prod A, <_{J^{\mathrm{bd}}_{\lambda}} \right) \leq \mathsf{pp}(\lambda) \right\}.$$

Then $\mathcal{I}$ is a possibly improper ideal on $\lambda \cap \mathrm{Reg}$, containing every bounded subset.

Why $\mathcal{I}$ Is Proper

Our standing assumption is

$$\mathsf{pp}(\lambda) < \mathsf{cov}(\lambda, \lambda, \aleph_1, 2).$$

By our earlier reduction,

$$\begin{aligned}\mathsf{pp}(\lambda) &< \mathsf{cf}_{\aleph_0}\left(\prod(\lambda \cap \mathrm{Reg}), <_{J^{\mathrm{bd}}_\lambda}\right) \\ &\leq \mathsf{cf}_{\mathfrak{c}}\left(\prod(\lambda \cap \mathrm{Reg}), <_{J^{\mathrm{bd}}_\lambda}\right).\end{aligned}$$

Hence $\lambda \cap \mathrm{Reg} \notin \mathcal{I}$, so $\mathcal{I}$ is proper.

Defining H_{bd}

Fix increasing cardinals

$$\mathfrak{c}^+ < \lambda_0 < \lambda_1 < \cdots, \qquad \sup_{n<\omega} \lambda_n = \lambda.$$

For cardinal sequences $\bar{\mu} = \langle \mu_n : n < \omega \rangle$, put

$$H = \{\bar{\mu} : \lambda_n \leq \mu_n \leq \lambda_{n+1} \text{ for all } n < \omega\}.$$

For $\bar{\mu} \in H$, let

$$R_{\bar{\mu}} = \bigcup_{n<\omega} \big(\mathrm{Reg} \cap [\lambda_n, \mu_n)\big).$$

Then define

$$H_{\mathrm{bd}} = \{\bar{\mu} \in H : R_{\bar{\mu}} \in \mathcal{I}\}.$$

These are *good* sequences for the remainder of the talk.

Directedness of H_{bd}

If

$$\mathsf{cov}\big(\mathsf{pp}(\lambda), \lambda^+, \mathfrak{c}^+, 2\big) \leq \mathsf{pp}(\lambda),$$

then

$$(H_{\mathrm{bd}}, \leq^*) \text{ is } \mathfrak{c}^+\text{-directed.}$$

Every family of at most $\mathfrak{c}$ good sequences has a good upper bound (modulo J^{bd}_ω)

Upper bounds at limits

Let $\langle \bar{\mu}^\xi : \xi < \delta \rangle$ be a $\leq^*$-increasing sequence in $H_{\rm bd}$, where

$$\mathrm{cf}(\delta) > \mathfrak{c} \qquad \text{and} \qquad \delta < \mathrm{pp}(\lambda)^+.$$

Then it has a good eventual upper bound:

$$\bar{\mu} \in H_{\rm bd}, \qquad \bar{\mu}^\xi \leq^* \bar{\mu} \quad (\xi < \delta).$$

PCF theory supplies an exact upper bound in H. Such an exact upper bound is again good.

Increasing a good sequence

Given $\bar{\mu} = \langle \mu_n : n < \omega \rangle \in H_{\mathrm{bd}}$, set

$$\mu'_n = \mathrm{min}\{\mu_n^+, \lambda_{n+1}\}, \qquad \bar{\mu}' = \langle \mu'_n : n < \omega \rangle.$$

Then

$$\bar{\mu}' \in H_{\mathrm{bd}} \qquad \text{and} \qquad \bar{\mu} \leq^* \bar{\mu}'.$$

Unless $\bar{\mu} =^* \langle \lambda_{n+1} : n < \omega \rangle$, we have

$$\bar{\mu} \neq^* \bar{\mu}'.$$

The first stopping stage

Assume $\chi = \lambda$. Build a $\leq^*$-increasing chain in H_{bd}, with

$$\bar{\mu}^0 = \langle \lambda_n : n < \omega \rangle, \qquad \bar{\mu}^{\xi+1} \neq^* \bar{\mu}^\xi.$$

Since H is a set, there is a first stopping stage δ.

If δ is a limit:

- $\mathrm{cf}(\delta) \leq \mathfrak{c}$: directedness gives a good upper bound.
- $\mathrm{cf}(\delta) > \mathfrak{c}$: pcf theory provides an exact upper bound which is again good.

Reaching the top

Thus $\delta = \gamma + 1$. Successor closure forces

$$\bar{\mu}^\gamma =^* \langle \lambda_{n+1} : n < \omega \rangle.$$

Since $\bar{\mu}^\gamma$ is good, $R_{\bar{\mu}^\gamma} \in \mathcal{I}$ contains a tail of $\lambda \cap \mathrm{Reg}$.

The ideal $\mathcal{I}$ contains bounded sets, so

$$\lambda \cap \mathrm{Reg} \in \mathcal{I},$$

contradicting properness. Therefore

$$\chi > \lambda.$$